

IE 203 - Operations Research II
Quiz II - Solutions

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March 06, 2026

Duration: 45 minutes
This is a CLOSED BOOK exam.

Question I (60 pts.)

a. The knapsack formulation:

$$\begin{aligned} \max \quad & z = 16x_1 + 22x_2 + 12x_3 + 8x_4 \\ \text{s.t.} \quad & 5x_1 + 7x_2 + 4x_3 + 3x_4 \leq 14 \\ & x_j = 0, 1 \quad j = 1, 2, 3, 4 \end{aligned}$$

b. Branch and bound solution

The method used in traversing the tree of Figure 1 is as follows:

1. LIFO approach to determine which subproblem should be solved.
2. Subproblem 3 is arbitrarily chosen before subproblem 2. To solve subproblem 3, we first set $x_3 = 1$ and then solved the resulting knapsack problem. After setting $x_3 = 1$, $14 - 4 = 10$ million TL was still available for investment. Applying the technique used to solve the LP relaxation of a knapsack problem yielded the following optimal solution to subproblem 3: $x_3 = 1$, $x_1 = 1$, $x_2 = 5/7$, $x_4 = 0$, $z = 16 + 12 + (5/7)22 = 306/7$. Other subproblems were solved similarly; if a subproblem specified $x_i = 0$, the optimal solution to that subproblem could not use investment i .
3. Subproblem 4 yielded the candidate solution $x_1 = x_2 = x_3 = 1$, $z = 16$. We then set $LB = 36$.
4. Subproblem 6 yielded a candidate solution with $z = 42$. Thus, subproblem 4 was eliminated from consideration, and the LB was updated to 42.
5. Subproblem 7 was infeasible because it required $x_1 = x_2 = x_3 = 1$, and such solution requires at least 16 million TL.
6. Subproblem 8 was eliminated because its z -value ($z = 38$) did not exceed the current LB of 42.
7. Subproblem 9 had a z -value of $300/7$. Because the z -value for any all-integer solution must also be an integer, this meant that branching on subproblem 9 could never yield a z -value larger than 42. Thus, further branching on subproblem 9 could not beat the current LB of 42, and subproblem 9 was eliminated from consideration.

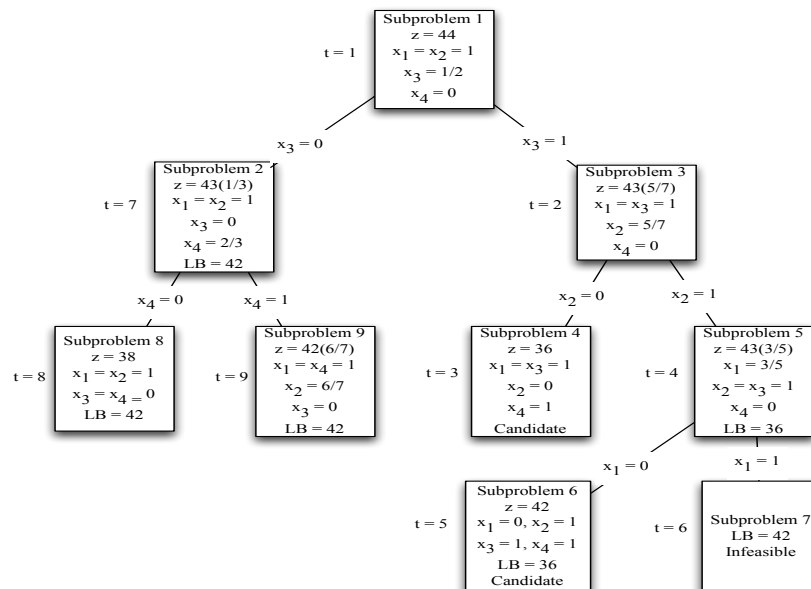


Figure 1: Branch-and-bound search tree

Question II (40 pts.)

The Lagrangean function can be written as

$$\mathcal{L}(x_1, x_2, \lambda, \mu_1, \mu_2) = (2x_1 - 1)^2 + x_2^2 + \lambda(-2x_1^2 + x_2^2) + \mu_1(-x_1 - 1) + \mu_2(-x_2 - 2).$$

Then, KKT first order conditions becomes,

$$\begin{aligned} 4(2x_1 - 1) - 4\lambda x_1 - \mu_1 &= 0 \\ 2x_2 + 2\lambda x_2 - \mu_2 &= 0 \\ \mu_1(-x_1 - 1) &= 0 \\ \mu_2(-x_2 - 2) &= 0 \\ \mu_1, \mu_2 &\geq 0 \end{aligned}$$

Consider the case with the last two inequalities are not active at $\mathbf{x}^*$. Then $\mu_1 = \mu_2 = 0$. By the second equality of the KKT conditions

$$2(1 + \lambda)x_2 = 0,$$

and hence there are two cases. In one of them, $x_2 = 0$. Then by the constraint $-2x_1^2 + x_2^2 = 0$, $x_1 = 0$, and by the first equality $\mu_1 = -4$, which is not possible. The second case is when $\lambda = -1$, and then by the first equality $4(2x_1 - 1) = -4$, that is, $x_1^* = \frac{1}{3}$, and then $x_2^* = \pm \frac{\sqrt{2}}{3}$. We thus obtain two stationary points: $x_1^* = \frac{1}{3}$, $x_2^* = \frac{\sqrt{2}}{3}$ with multipliers $\lambda^* = -1$ and $\mu_1^* = \mu_2^* = 0$; $x_1^* = \frac{1}{3}$, $x_2^* = -\frac{\sqrt{2}}{3}$ with multipliers $\lambda^* = -1$ and $\mu_1^* = \mu_2^* = 0$. Notice that these two stationary points are regular points.